

Implicit Meta-Attention

- Context of contexts

Gary Nan Tie, Aug 28, 2026

Abstract

Attention works because token context is taken into account. Implicit deep learning models of attention are composable resulting in a context of contexts, meta-attention so to speak. This mechanism potentially makes LLM more robust against drift and hallucinations.

Implicit Meta-Attention

Suppose $t_1, t_2, \dots, t_i$ is a token string

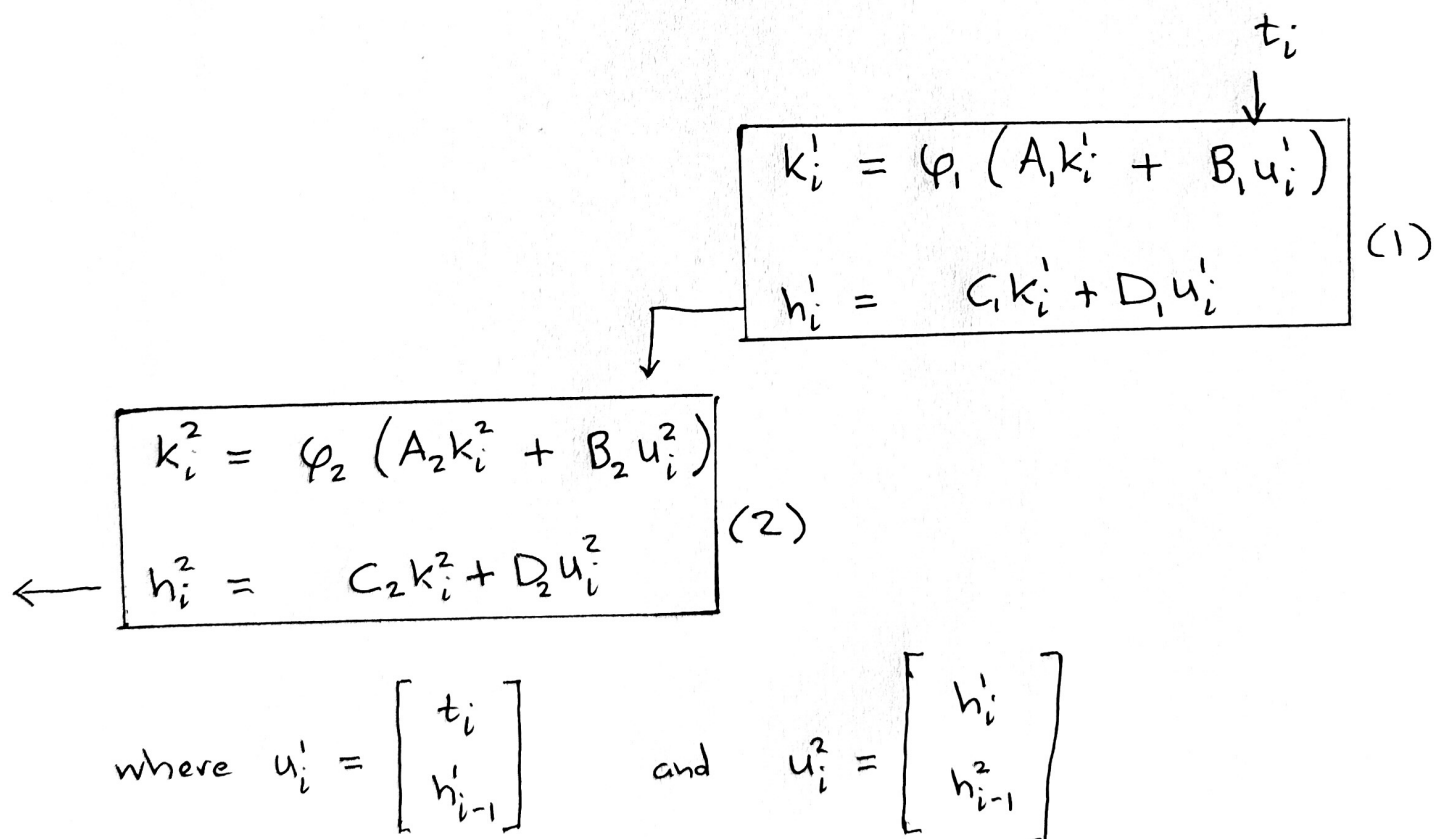
that we wish to find a low perplexity continuation.

State machine $\text{Alice}(\$, \mathcal{O})$ - see Appendix,

with output $o_{i+1} = \mathcal{O}(h_i)$ where $h_i = \$ (h_{i-1}, t_i)$,

gives a continuation $t_1, \dots, t_i, o_{i+1}$

Now implicit deep learning models can be composed:



In (1), h_i^1 is a contextual representation of token t_i .

In (2), h_i^2 is a contextual representation of state h_i^1 .

In this way, the composition of implicit deep learning models is an attention hierarchy, a context of contexts.

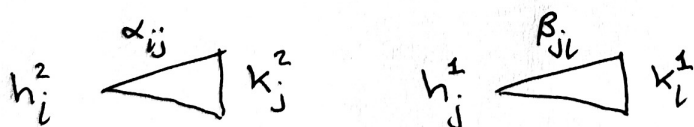
Train (1) to learn state machine Alice ($\mathbb{S}_0, \mathbb{Q}_1$)

(see Appendix). In particular, we know a

distribution β_{jl} , of h_j^1 over k_l^1 , with $l_j^\beta \triangleq \arg \max_{l \in [J]} \beta_{jl}$.

Let $\alpha_{ij} \triangleq \text{softmax score}(h_i^2, k_j^2)$

and let $j_i^\alpha \triangleq \arg \max_{j \in [I]} \alpha_{ij}$.



Define the output of (2) to be:

$$o_{i+1} \triangleq t_{l_j^\beta(j_i^\alpha)}$$

In words, given h_i^2 we first use α_{ij} to choose $h_{j_i^\alpha}^1$. Let $j^* = j_i^\alpha$. Now use β_{j^*l} to choose $t_{l_{j^*}}^\beta$.

So we have a state machine $(\mathcal{S}_2, \mathcal{O}_2)$ corresponding to (2), where:

$$h_i^2 = \mathcal{S}_2(h_{i-1}^2, h_i^1) = C_2 k_i^2 + D_2 u_i^2$$

$$\text{and } o_{i+1} = \mathcal{O}_2(h_i^2) = t_{l_{j^*}}^\beta, \quad j^* = j_i^\alpha.$$

Implicit Meta-Attention Mechanism

In the context of the composition of implicit deep learning models (1) and (2), first learn state machine Alice $(\mathcal{S}_1, \mathcal{O}_1)$ from (1).

From (2) learn state machine $(\mathcal{S}_2, \mathcal{O}_2)$, that is learn matrices A_2, B_2, C_2, D_2 and fixed points k_j^2 to minimize Perplexity $(o_2, \dots, o_{m+1})$.

Appendix

State Machine $(\mathcal{S}, \mathcal{O})$

Input: tokens $t_1, \dots, t_m \in \mathbb{R}^p$

Initial state: $h_0 = 0 \in \mathbb{R}^q$

Next state: $h_i = \mathcal{S}(h_{i-1}, t_i) \in \mathbb{R}^q, i \in [m]$

Output: $o_{i+1} = \mathcal{O}(h_i) \in \mathbb{R}^p, i \in [m]$

where $\mathcal{S}$ is a transition function and $\mathcal{O}$ an output function.

Let $\text{Emb}: \mathbb{R}^p \rightarrow \mathbb{R}^{p'}$ be an embedding

$$t_i \mapsto v_i, i \in [m]$$

Let $\Theta = \{(W_1, b_1), (W_2, b_2)\}$ be parameters for
a feedforward network defined by:

$$\text{FFN}_{\Theta}(h_i) \triangleq \text{ReLU}(h_i W_1 + b_1) W_2 + b_2$$

implicit RNN

$$h_0 = 0 \in \mathbb{R}^q, \quad k_0 = 0 \in \mathbb{R}^q$$

$$u_i = \begin{pmatrix} t_i \\ h_{i-1} \end{pmatrix} \in \mathbb{R}^r, \quad r = p + q$$

$$k_i = \varphi(Ak_i + Bu_i) \in \mathbb{R}^q$$

$$h_i = \mathcal{S}(h_{i-1}, t_i) \triangleq Ck_i + Du_i \in \mathbb{R}^q$$

where $A \triangleq \begin{bmatrix} I_p & 0 \\ 0 & \gamma I_q \end{bmatrix} + \gamma Id_q$, $0 < \gamma < 1$,

ensures A is well-posed for φ , that is,

the Fixed point equation has a solution and it is unique.

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So an implicit RNN constructs an $\mathcal{S}$.

Attention Layer

Input h_i

Let score be a similarity function eg a reproducing kernel.

For $i \in [m]$, define probabilities

$$\alpha_{ij} \triangleq \text{softmax}_{j \in [i]} \text{score}(h_i, k_j)$$

Let $\hat{h}_i = \sum_{j=1}^i \alpha_{ij} v_j$, where $v_j = \text{Emb}(t_j)$

and define update $h'_i = \text{FFN}_{\theta}(\hat{h}_i)$. ($\dim h'_i = \dim k_j$)

Output h'_i

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Stack L attention layers $\Theta \triangleq \{\theta_1, \dots, \theta_L\}$

composing updates FFN_{θ_L} , $L \in [L]$,

and now let h'_i denote the final L -th update.

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After L layers, for $i \in [m]$ define probabilities

$$\beta_{ij} \triangleq \text{softmax}_{j \in [i]} \text{score}(h'_i, k_j) \text{ and}$$

$$\text{let } j_i^B \triangleq \arg \max_{j \in [i]} \beta_{ij}$$

$$\text{Define token continuation } o_{i+1} = \Phi(h_i) \triangleq t_{j_i^B}$$

So attention layers construct an $\oplus$

Define the average Renyi α -entropy of $\{t_1, \dots, t_m\}$ by:

$$E(\{t_i\}_{i=1}^m) \triangleq \frac{1}{m} \sum_{i=1}^m H_\alpha^i$$

$$\text{where } H_\alpha^i \triangleq H_\alpha(\{\beta_{i,j}\}_{j=1}^i)$$

and H_α is Renyi α -entropy.

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Token continuation $t_1, t_2, \dots, t_i, o_{i+1}$ via:

Alice($\mathcal{S}, \mathcal{O}$) training problem (next-token)

Let $\lambda > 0$, learn implicit RNN matrices A, B, C, D

and fixed points k_i , as well as feedforward network

parameters Θ to minimize:

$$\left[\text{Perplexity}(o_2, \dots, o_{mn}) + \lambda E(\{t_i\}_{i=1}^m) \right]$$

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